

Luca Fanucci

professor

基本信息

姓名: Luca Fanucci

所属院校: 比萨大学

地区: Italy

年龄范围: 30岁至40岁

海外经历: 有

学科领域

电子与计算机工程

通信工程

人工智能

航空航天工程

微电子科学与工程

集成电路设计与集成系统

信息安全

研究领域

航电系统与机载设备

FPGA与可重构计算

卫星通信与空间通信

深度学习与神经网络

密码学与密码分析

片上系统与嵌入式设计

智能边缘计算

嵌入式系统与硬件设计

超大规模集成电路设计

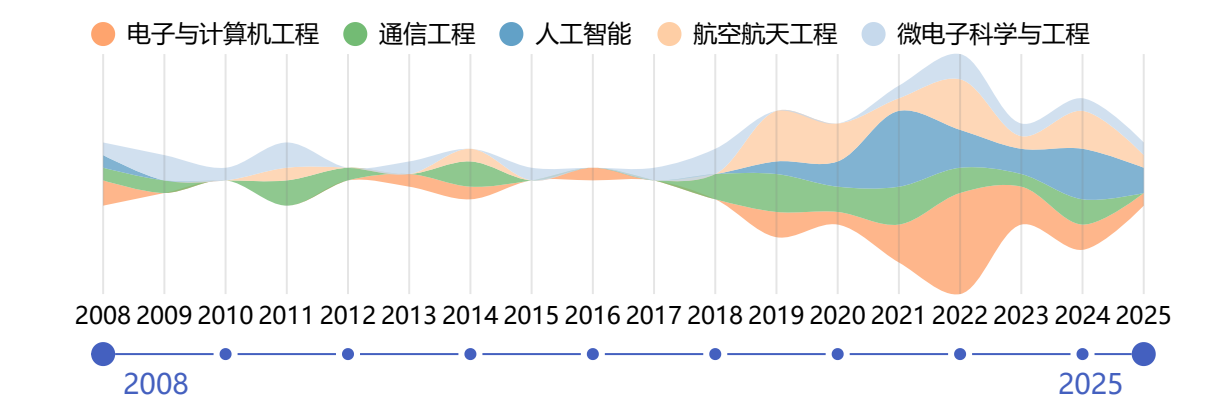
通信系统仿真与优化

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个人简介

Luca Fanucci is a professor of Electronics at the Department of Information Engineering, University of Pisa. He graduated cum laude in Electronic Engineering in 1992 and obtained his Ph.D. in Information Engineering in 1996 from the same university. He has held various research and academic positions since then. Since 2011, he has collaborated on supporting students with disabilities. He is also involved in institutional roles, teaching, and research. His research focuses on integrated circuits and embedded systems. He has over 400 publications and 40+ patents. He has led numerous projects and served on technical committees. He is an associate editor of several journals.

近年发表的论文



Hardware Design of an Advanced-Feature Cryptographic Tile Within the European Processor Initiative

IEEE TRANSACTIONS ON COMPUTERS

Nannipieri, Pietro; Crocetti, Luca; Di Matteo, Stefano; Fanucci, Luca; Saponara, Sergio

被引用次数: 8 2025

RADSAFiE: A Netlist-Level Fault Injection User Interface Application for FPGA-Based Digital Systems

IEEE ACCESS

Monopoli, Matteo; Biondi, Matteo; Nannipieri, Pietro; Moranti, Silvia; Fanucci, Luca

被引用次数: 0 2025

Medical Expert System for Intelligent Telemonitoring of Patients With Chronic Heart Failure: Preliminary Validation and Perspectives

CIRCULATION-HEART FAILURE

Vianello, Annamaria; Olivelli, Martina; Donati, Massimiliano; Fanucci, Luca; Bechini, Alessio; Petrucci, Ilaria; Masi, Stefano

被引用次数: 0 2025

Efficient Coarse-Grained Reconfigurable Array architecture for machine learning applications in space using DARE65T library platform

MICROPROCESSORS AND MICROSYSTEMS

Zulberti, Luca; Monopoli, Matteo; Nannipieri, Pietro; Moranti, Silvia; Thys, Geert; Fanucci, Luca

被引用次数: 0 2025

Integration of Twin Models in UVM Verification IPs for Space Telecommunication Systems

IEEE ACCESS

Vagaggini, Simone; Davalle, Daniele; Nannipieri, Pietro; Fanucci, Luca

被引用次数: 0 2024

Improving the Efficiency of Software-Based Fault Protection Mechanisms With HUSTLE

IEEE ACCESS

Ferrante, Nicola; Fanucci, Luca; Rossi, Francesco; Terrosi, Francesco; Bondavalli, Andrea

被引用次数: 0

2024

Design and Implementation of a Configurable Fully Compliant DVB-S2 LDPC Encoder for High Data-Rate Downlink Payload

IEEE ACCESS

Nannipieri, Pietro; Bartolacci, Giacomo; Bertolucci, Matteo; Fanucci, Luca

被引用次数: 0

2024

GPU@SAT DevKit: Empowering Edge Computing Development Onboard Satellites in the Space-IoT Era

ELECTRONICS

Benelli, Gionata; Todaro, Giovanni; Monopoli, Matteo; Giuffrida, Gianluca; Donati, Massimiliano; Fanucci, Luca

被引用次数: 1

2024

Design and Evaluation of CPU-, GPU-, and FPGA-Based Deployment of a CNN for Motor Imagery Classification in Brain-Computer Interfaces

ELECTRONICS

Pacini, Federico; Pacini, Tommaso; Lai, Giuseppe; Zocco, Alessandro Michele; Fanucci, Luca

被引用次数: 2

2024

Enhancing Precision of Telemonitoring of COVID-19 Patients through Expert System Based on IoT Data Elaboration

ELECTRONICS

Olivelli, Martina; Donati, Massimiliano; Vianello, Annamaria; Petrucci, Ilaria; Masi, Stefano; Bechini, Alessio; Fanucci, Luca

被引用次数: 0

2024

Design of an Assisted Driving System for Obstacle Avoidance Based on Reinforcement Learning Applied to Electrified Wheelchairs

ELECTRONICS

Pacini, Federico; Dini, Pierpaolo; Fanucci, Luca

被引用次数: 4

2024

Scalable Hardware-Efficient Architecture for Frame Synchronization in High-Data-Rate Satellite Receivers

ELECTRONICS

Crocetti, Luca; Pagani, Emanuele; Bertolucci, Matteo; Fanucci, Luca

被引用次数: 0

2024

Show Me Once: A Transformer-Based Approach for an Assisted-Driving System

MACHINE LEARNING AND KNOWLEDGE EXTRACTION

Pacini, Federico; Dini, Pierpaolo; Fanucci, Luca

被引用次数: 0

2024

Design and Validation of an Intelligent System for Decision-making on Telemonitoring Intensity in Heart Failure Patients

2024 18TH INTERNATIONAL SYMPOSIUM ON MEDICAL INFORMATION AND COMMUNICATION
TECHNOLOGY, ISMICT 2024

Olivelli, Martina; Donati, Massimiliano; Vianello, Annamaria; Petrucci, Ilaria; Masi, Stefano; Bechini, Alessio;
Fanucci, Luca

被引用次数: 0

2024

A Scalable and Configurable Architecture for Hardware Authenticated Encryption Modules Compliant with the CCSDS Security Specifications

2024 SECURITY FOR SPACE SYSTEMS CONFERENCE, 3S 2024

Crocetti, Luca; Fanucci, Luca; Baldanzi, Luca; Falaschi, Francesco

被引用次数: 0

2024

Design and Preliminary Validation of an Assisted Driving System for Obstacle Avoidance Based on Reinforcement Learning Applied to Electrified Wheelchairs

COMPUTERS HELPING PEOPLE WITH SPECIAL NEEDS, PT II, ICCHP 2024

Pacini, Federico; Fanucci, Luca

被引用次数: 0

2024

Innovative Plug-and-Play System for Electrification of Wheel-Chairs

IEEE ACCESS

Pacini, Federico; Matteo, Stefano Di; Dini, Pierpaolo; Fanucci, Luca; Bucchi, Francesco

被引用次数: 9

2023

FPG-AI: A Technology-Independent Framework for the Automation of CNN Deployment on FPGAs

IEEE ACCESS

Pacini, Tommaso; Rapuano, Emilio; Fanucci, Luca

被引用次数: 3

2023

Review of Methodologies and Metrics for Assessing the Quality of Random Number Generators

ELECTRONICS

Crocetti, Luca; Nannipieri, Pietro; Di Matteo, Stefano; Fanucci, Luca; Saponara, Sergio

被引用次数: 9

2023

ECG-Based Stress Detection and Productivity Factors Monitoring: The Real-Time Production Factory System

SENSORS

Donati, Mssimiliano; Olivelli, Martina; Giovannini, Romano; Fanucci, Luca

被引用次数: 4

2023

SpaceART SpaceWire Sniffer for Link Monitoring: A Complete Communication Analysis in a Time-Constrained Test Scenario

SENSORS

Ciardi, Roberto; Vagaggini, Simone; Marino, Antonino; Nannipieri, Pietro; Fanucci, Luca

被引用次数: 0

2023

Remotely Controlled Electronic Goalkeeper: An Example of Improving Social Integration of Persons with and without Disabilities

APPLIED SCIENCES-BASEL

Donati, Massimiliano; Pacini, Federico; Baldanzi, Luca; Turturici, Mauro; Fanucci, Luca

被引用次数: 2

2023

Implementation of a Hash-based Secure Core for Integrity and Authentication of Data in Space Applications on Space-grade FPGAs

2023 IEEE 2ND INDUSTRIAL ELECTRONICS SOCIETY ANNUAL ON-LINE CONFERENCE, ONCON

Crocetti, Luca; Falaschi, Francesco; Saponara, Sergio; Fanucci, Luca

被引用次数: 1

2023

Implementation Strategies for Highly-accurate and Efficient Frame Synchronization Modules in Satellite Communication Receivers

2023 IEEE 2ND INDUSTRIAL ELECTRONICS SOCIETY ANNUAL ON-LINE CONFERENCE, ONCON

Crocetti, Luca; Pagani, Emanuele; Bertolucci, Matteo; Fanucci, Luca

被引用次数: 0

2023

The Φ -Sat-1 Mission: The First On-Board Deep Neural Network Demonstrator for Satellite Earth Observation

IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING

Giuffrida, Gianluca; Fanucci, Luca; Meoni, Gabriele; Batic, Matej; Buckley, Leonie; Dunne, Aubrey; van Dijk, Chris; Esposito, Marco; Hefele, John; Vercruyssen, Nathan; Furano, Gianluca; Pastena, Massimiliano; Aschbacher, Josef

被引用次数: 77

2022

Design and Test of an Integrated Random Number Generator with All-Digital Entropy Source

ENTROPY

Crocetti, Luca; Di Matteo, Stefano; Nannipieri, Pietro; Fanucci, Luca; Saponara, Sergio

被引用次数: 11

2022

VLSI Design of Advanced-Features AES Cryptoprocessor in the Framework of the European Processor Initiative

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Nannipieri, Pietro; Matteo, Stefano Di; Baldanzi, Luca; Crocetti, Luca; Zulberti, Luca; Saponara, Sergio; Fanucci, Luca

被引用次数: 11

2022

A Script-Based Cycle-True Verification Framework to Speed-Up Hardware and Software Co-Design: Performance Evaluation on ECC Accelerator Use-Case

ELECTRONICS

Zulberti, Luca; Di Matteo, Stefano; Nannipieri, Pietro; Saponara, Sergio; Fanucci, Luca

被引用次数: 9

2022

A 60GS/s Two-Stage Sampler with a Linearity Calibration Loop for PAM-8 Receivers

ELECTRONICS

Di Pasquo, Alessio; Monaco, Enrico; Ghittori, Nicola; Nani, Claudio; Fanucci, Luca

被引用次数: 0

2022

A Track-and-Hold Circuit with Tunable Non-Linearity and a Calibration Loop for PAM-8 SerDes Receivers

ELECTRONICS

Di Pasquo, Alessio; Monaco, Enrico; Ghittori, Nicola; Nani, Claudio; Fanucci, Luca

被引用次数: 2

2022

Industrial functional safety assessment for WSN using QoS metrics

HELIYON

Srinivasan, Sivasubramanian; Ramesh, T. K.; Paccapeli, Roberto; Fanucci, Luca

被引用次数: 1

2022

RETRACTED: A Post-training Quantization Method for the Design of Fixed-Point-Based FPGA/ASIC Hardware Accelerators for LSTM/GRU Algorithms (Retracted Article)

COMPUTATIONAL INTELLIGENCE AND NEUROSCIENCE

Rapuano, Emilio; Pacini, Tommaso; Fanucci, Luca

被引用次数: 4

2022

Architectural Implications for Inference of Graph Neural Networks on CGRA-based Accelerators

PRIME 2022: 17TH INTERNATIONAL CONFERENCE ON PHD RESEARCH IN MICROELECTRONICS AND ELECTRONICS

Zulberti, Luca; Monopoli, Matteo; Nannipieri, Pietro; Fanucci, Luca

被引用次数: 5

2022

CCSDS 131.2-B-1 Software Defined Radio receiver featuring GPU accelerators: up to 1000x with respect to CPU implementation

2022 9TH INTERNATIONAL WORKSHOP ON TRACKING, TELEMETRY AND COMMAND SYSTEMS FOR SPACE APPLICATIONS (TTC)

Ciardi, Roberto; Giuffrida, Gianluca; Bertolucci, Matteo; Pagani, Emanuele; Fanucci, Luca

被引用次数: 1

2022

A CCSDS 131.2-B compliant telemetry downlink hardware-software emulation platform for next-generation Earth observation missions

2022 9TH INTERNATIONAL WORKSHOP ON TRACKING, TELEMETRY AND COMMAND SYSTEMS FOR SPACE APPLICATIONS (TTC)

Pagani, Emanuele; Ciardi, Roberto; Benelli, Gionata; Bertolucci, Matteo; Falaschi, Francesco; Cassettari, Riccardo; Fanucci, Luca; Modenini, Andrea

被引用次数: 1

2022

SpaceFibre Network Discovery and Configuration Protocol Development

SPACEWIRE AND SPACEFIBRE 2022: PROCEEDINGS OF THE 9TH 2022 INTERNATIONAL SPACEWIRE & SPACEFIBRE CONFERENCE (ISC)

Nannipieri, Pietro; Benelli, Gionata; Davalle, Daniele; Fanucci, Luca; Jameux, David

Definition, Implementation and Testing of an XML-based Packet Description Language for Traffic Analysis in a SpaceWire Communication

SPACEWIRE AND SPACEFIBRE 2022: PROCEEDINGS OF THE 9TH 2022 INTERNATIONAL SPACEWIRE & SPACEFIBRE CONFERENCE (ISC)

Ciardi, Roberto; Vagaggini, Simone; Marino, Antonino; Davalle, Daniele; Benelli, Gionata; Fanucci, Luca

被引用次数: 0

2022

SpaceWire Codec VIP: an innovative architecture of UVM-based Verification Environment

SPACEWIRE AND SPACEFIBRE 2022: PROCEEDINGS OF THE 9TH 2022 INTERNATIONAL SPACEWIRE & SPACEFIBRE CONFERENCE (ISC)

Vagaggini, Simone; Trafeli, Marco; Ciardi, Roberto; Davalle, Daniele; Santos, Lucana; Nannipieri, Pietro; Fanucci, Luca

被引用次数: 0

2022

SystemVerilog UVM-based Verification Environment for a SpaceFibre Router

SPACEWIRE AND SPACEFIBRE 2022: PROCEEDINGS OF THE 9TH 2022 INTERNATIONAL SPACEWIRE & SPACEFIBRE CONFERENCE (ISC)

Gigli, Lorenzo; Nannipieri, Pietro; Zulberti, Luca; Vagaggini, Simone; Fanucci, Luca

被引用次数: 0

2022

Oil Spill Identification from SAR Images for Low Power Embedded Systems Using CNN

REMOTE SENSING

Diana, Lorenzo; Xu, Jia; Fanucci, Luca

被引用次数: 13

2021

An FPGA-Based Hardware Accelerator for CNNs Inference on Board Satellites: Benchmarking with Myriad 2-Based Solution for the CloudScout Case Study

REMOTE SENSING

Rapuano, Emilio; Meoni, Gabriele; Pacini, Tommaso; Dinelli, Gianmarco; Furano, Gianluca; Giuffrida, Gianluca; Fanucci, Luca

被引用次数: 32

2021

The very high-speed SpaceFibre multi-lane CoDec: Implementation and experimental performance evaluation

ACTA ASTRONAUTICA

Dinelli, Gianmarco; Nannipieri, Pietro; Marino, Antonino; Fanucci, Luca; Dello Sterpaio, Luca

被引用次数: 5

2021

Machine learning assistive application for users with speech disorders

APPLIED SOFT COMPUTING

Mulfari, Davide; Meoni, Gabriele; Marini, Marco; Fanucci, Luca

被引用次数: 22

2021

A RISC-V Post Quantum Cryptography Instruction Set Extension for Number Theoretic Transform to Speed-Up CRYSTALS Algorithms

IEEE ACCESS

Nannipieri, Pietro; Di Matteo, Stefano; Zulberti, Luca; Albicocchi, Francesco; Saponara, Sergio; Fanucci, Luca

被引用次数: 36

2021

SHA2 and SHA-3 accelerator design in a 7 nm technology within the European Processor Initiative

MICROPROCESSORS AND MICROSYSTEMS

Nannipieri, Pietro; Bertolucci, Matteo; Baldanzi, Luca; Crocetti, Luca; Di Matteo, Stefano; Falaschi, Francesco; Fanucci, Luca; Saponara, Sergio

被引用次数: 15

2021

A Multi-Cache System for On-Chip Memory Optimization in FPGA-Based CNN Accelerators

ELECTRONICS

Pacini, Tommaso; Rapuano, Emilio; Dinelli, Gianmarco; Fanucci, Luca

被引用次数: 2

2021

Secure Elliptic Curve Crypto-Processor for Real-Time IoT Applications

ENERGIES

Di Matteo, Stefano; Baldanzi, Luca; Crocetti, Luca; Nannipieri, Pietro; Fanucci, Luca; Saponara, Sergio

被引用次数: 26

2021

Telemonitoring in the Covid-19 Era: The Tuscany Region Experience

HEALTHCARE

Panicacci, Silvia; Donati, Massimiliano; Lubrano, Alberto; Vianello, Annamaria; Ruiiu, Alessio; Melani, Luca; Tomei, Antonella; Fanucci, Luca

被引用次数: 12

2021

Optimising Speaker-Dependent Feature Extraction Parameters to Improve Automatic Speech Recognition Performance for People with Dysarthria

SENSORS

Marini, Marco; Vanello, Nicola; Fanucci, Luca

被引用次数: 5

2021

On the Frequency Carrier Offset and Symbol Timing Estimation for CCSDS 131.2-B-1 High Data-Rate Telemetry Receivers

SENSORS

Bertolucci, Matteo; Cassettari, Riccardo; Fanucci, Luca

被引用次数: 5

2021

An Easily Integrable Industrial System for Gamma Spectroscopic Analysis and Traceability of Stones and Building Materials

SENSORS

Marini, Marco; Panicacci, Silvia; Donati, Massimiliano; Fanucci, Luca; Fanchini, Erica; Pepperosa, Andrea; Morichi, Massimo; Alberi, Matteo; Chiarelli, Enrico; Montuschi, Michele; Raptis, Kassandra Giulia Cristina; Serafini, Andrea; Strati, Virginia; Mantovani, Fabio

被引用次数: 3	2021
Data-Driven Convolutional Model for Digital Color Image Demosaicing	
APPLIED SCIENCES-BASEL	
de Gioia, Francesco; Fanucci, Luca	
被引用次数: 1	2021
True Random Number Generator Based on Fibonacci-Galois Ring Oscillators for FPGA	
APPLIED SCIENCES-BASEL	
Nannipieri, Pietro; Di Matteo, Stefano; Baldanzi, Luca; Crocetti, Luca; Belli, Jacopo; Fanucci, Luca; Saponara, Sergio	
被引用次数: 24	2021
Empowering Home Health Monitoring of Covid-19 Patients with Smartwatch Position and Fitness Tracking	
2021 IEEE 34TH INTERNATIONAL SYMPOSIUM ON COMPUTER-BASED MEDICAL SYSTEMS (CBMS)	
Panicacci, Silvia; Giuffrida, Gianluca; Donati, Massimiliano; Lubrano, Alberto; Ruii, Alessio; Fanucci, Luca	
被引用次数: 9	2021
Functional Safety Integrity Assessment for Industrial Wireless Sensor Network Using QoS Metrics	
2021 IEEE 3RD PHD COLLOQUIUM ON ETHICALLY DRIVEN INNOVATION AND TECHNOLOGY FOR SOCIETY (PHD EDITS)	
Srinivasan, Sivasubramanian; Ramesh, T. K.; Paccapeli, Roberto; Fanucci, Luca	
被引用次数: 1	2021
Delay-lines jitter modeling and efficiency analysis in FinFET technology	
2021 28TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS, AND SYSTEMS (IEEE ICECS 2021)	
Di Pasquo, Alessio; Monaco, Enrico; Nani, Claudio; Fanucci, Luca	
被引用次数: 1	2021
Highly Parallel Sample Rate Converter for Space Telemetry Transmitters	
2021 24TH EUROMICRO CONFERENCE ON DIGITAL SYSTEM DESIGN (DSD 2021)	
Bertolucci, Matteo; Fanucci, Luca	
被引用次数: 1	2021
CloudScout: A Deep Neural Network for On-Board Cloud Detection on Hyperspectral Images	
REMOTE SENSING	
Giuffrida, Gianluca; Diana, Lorenzo; de Gioia, Francesco; Benelli, Gionata; Meoni, Gabriele; Donati, Massimiliano; Fanucci, Luca	
被引用次数: 86	2020
A representative SpaceFibre network evaluation: Features, performances and future trends	
ACTA ASTRONAUTICA	
Nannipieri, Pietro; Fanucci, Luca; Siegle, Felix	

被引用次数: 6

2020

CCSDS 131.2-B-1 telemetry transmitter: A VHDL IP core and a validation architecture on board RTG4 FPGA

ACTA ASTRONAUTICA

Meoni, Gabriele; Cassettari, Riccardo; Bertolucci, Matteo; Marino, Antonino; Davalle, Daniele; Trafeli, Marco; Fanucci, Luca

被引用次数: 5

2020

A serial high-speed satellite communication CODEC: Design and implementation of a SpaceFibre interface

ACTA ASTRONAUTICA

Nannipieri, Pietro; Dinelli, Gianmarco; Marino, Antonino; Dello Sterpaio, Luca; Leoni, Alessandro; Fanucci, Luca; Davalle, Daniele

被引用次数: 11

2020

Analysis of sampling and noise error on the energy measurement of direct digitally acquired pulsed signals

NUCLEAR INSTRUMENTS & METHODS IN PHYSICS RESEARCH SECTION A-ACCELERATORS SPECTROMETERS DETECTORS AND ASSOCIATED EQUIPMENT

Degl'Innocenti, Irene; Boccardi, Andrea; Fanucci, Luca; Wendt, Manfred

被引用次数: 1

2020

MEM-OPT: A Scheduling and Data Re-Use System to Optimize On-Chip Memory Usage for CNNs On-Board FPGAs

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Dinelli, Gianmarco; Meoni, Gabriele; Rapuano, Emilio; Pacini, Tommaso; Fanucci, Luca

被引用次数: 12

2020

Towards the Use of Artificial Intelligence on the Edge in Space Systems: Challenges and Opportunities

IEEE AEROSPACE AND ELECTRONIC SYSTEMS MAGAZINE

Furano, Gianluca; Meoni, Gabriele; Dunne, Aubrey; Moloney, David; Ferlet-Cavrois, Veronique; Tavoularis, Antonis; Byrne, Jonathan; Buckley, Leonie; Psarakis, Mihalios; Voss, Kay-Obbe; Fanucci, Luca

被引用次数: 104

2020

A Robust RANSAC-Based Planet Radius Estimation for Onboard Visual Based Navigation SENSORS

de Gioia, Francesco; Meoni, Gabriele; Giuffrida, Gianluca; Donati, Massimiliano; Fanucci, Luca

被引用次数: 3

2020

Cryptographically Secure Pseudo-Random Number Generator IP-Core Based on SHA2 Algorithm

SENSORS

Baldanzi, Luca; Crocetti, Luca; Falaschi, Francesco; Bertolucci, Matteo; Belli, Jacopo; Fanucci, Luca; Saponara, Sergio

被引用次数: 25

2020

Clustering Algorithm for a Spaceborne Lightning Imager: Design, Trade-Off, and FPGA Implementation

IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING

Nannipieri, Pietro; Davalle, Daniele; Nencioni, Stefano; Lombardi, Paolo; Fanucci, Luca

被引用次数: 3

2019

VHDL Design of a SpaceFibre Routing Switch

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Leoni, Alessandro; Nannipieri, Pietro; Fanucci, Luca

被引用次数: 9

2019

A Configurable Hardware Word Re-Ordering Block for Multi-Lane Communication Protocols: Design and Use Case

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Nannipieri, Pietro; Dinelli, Gianmarco; Fanucci, Luca

被引用次数: 4

2019

Design of a Reduced SpaceFibre Interface: An Enabling Technology for Low-Cost Spacecraft High-Speed Data-Handling

AEROSPACE

Dinelli, Gianmarco; Nannipieri, Pietro; Davalle, Daniele; Fanucci, Luca

被引用次数: 8

2019

SHINe: Simulator for Satellite on-Board High-Speed Networks Featuring SpaceFibre and SpaceWire Protocols

AEROSPACE

Leoni, Alessandro; Nannipieri, Pietro; Davalle, Daniele; Fanucci, Luca; Jameux, David

被引用次数: 9

2019

A simulated approach to evaluate side-channel attack countermeasures for the Advanced Encryption Standard

INTEGRATION-THE VLSI JOURNAL

Crocetti, Luca; Baldanzi, Luca; Bertolucci, Matteo; Sarti, Luca; Carnevale, Bernardino; Fanucci, Luca

被引用次数: 13

2019

A Complete EGSE Solution for the SpaceWire and SpaceFibre Protocol Based on the PXI Industry Standard

SENSORS

Dello Sterpaio, Luca; Marino, Antonino; Nannipieri, Pietro; Dinelli, Gianmarco; Davalle, Daniele; Fanucci, Luca

被引用次数: 12

2019

A YOLOv2 Convolutional Neural Network-Based Human-Machine Interface for the Control of Assistive Robotic Manipulators

APPLIED SCIENCES-BASEL

Giuffrida, Gianluca; Meoni, Gabriele; Fanucci, Luca

被引用次数: 9

2019

A Novel Parallel 8B/10B Encoder: Architecture and Comparison with Classical Solution

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Nannipieri, Pietro; Davalle, Daniele; Fanucci, Luca

被引用次数: 6

2018

MACsec-Based Security for Automotive Ethernet Backbones

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Carnevale, Berardino; Fanucci, Luca; Bisase, Samson; Hunjan, Harman

被引用次数: 11

2018

Real-Time and High-Accuracy Arctangent Computation Using CORDIC and Fast Magnitude Estimation

ELECTRONICS

Pilato, Luca; Fanucci, Luca; Saponara, Sergio

被引用次数: 16

2017

Configurable network-on-chip router macrocells

MICROPROCESSORS AND MICROSYSTEMS

Saponara, Sergio; Fanucci, Luca

被引用次数: 3

2016

Predictive Diagnosis of High-Power Transformer Faults by Networking Vibration Measuring Nodes With Integrated Signal Processing

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Saponara, Sergio; Fanucci, Luca; Bernardo, Fabio; Falciani, Alessandro

被引用次数: 64

2016

An Embedded Sensing and Communication Platform, and a Healthcare Model for Remote Monitoring of Chronic Diseases

ELECTRONICS

Saponara, Sergio; Donati, Massimiliano; Fanucci, Luca; Celli, Alessio

被引用次数: 20

2016

Fully integrated multi-channel inductive load driver for harsh automotive applications

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Saponara, Sergio; Fanucci, Luca; Pasetti, Giuseppe; Serventi, Riccardo

被引用次数: 1

2016

A Multichannel and Compact Time to Digital Converter for Time of Flight Positron Emission Tomography

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Marino, Nahema; Baronti, Federico; Fanucci, Luca; Saponara, Sergio; Roncella, Roberto; Bisogni, Maria G.; Del Guerra, Alberto

被引用次数: 18

2015

Design of the 15 GHz BPM test bench for the CLIC test facility to perform precise stretched-wire RF measurements

MEASUREMENT SCIENCE AND TECHNOLOGY

Zorzetti, Silvia; Fanucci, Luca; Munoz, Natalia Galindo; Wendt, Manfred

被引用次数: 2

2015

Design of an NoC Interface Macrocell with Hardware Support of Advanced Networking Functionalities

IEEE TRANSACTIONS ON COMPUTERS

Saponara, Sergio; Bacchillone, Tony; Petri, Esa; Fanucci, Luca; Locatelli, Riccardo; Coppola, Marcello

被引用次数: 15

2014

Low-power DSP system for real-time correction of fish-eye cameras in automotive driver assistance applications

JOURNAL OF REAL-TIME IMAGE PROCESSING

Turturici, Mauro; Saponara, Sergio; Fanucci, Luca; Franchi, Emilio

被引用次数: 20

2014

DESIGN, IMPLEMENTATION AND TESTING OF A FLEXIBLE FULLY-DIGITAL TRANSPONDER FOR LOW-EARTH ORBIT SATELLITE COMMUNICATIONS

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Davalle, Daniele; Cassettari, Riccardo; Saponara, Sergio; Fanucci, Luca; Cucchi, Luca; Bigongiari, Franco; Errico, Walter

被引用次数: 5

2014

A multi-processor NoC-based architecture for real-time image/video enhancement

JOURNAL OF REAL-TIME IMAGE PROCESSING

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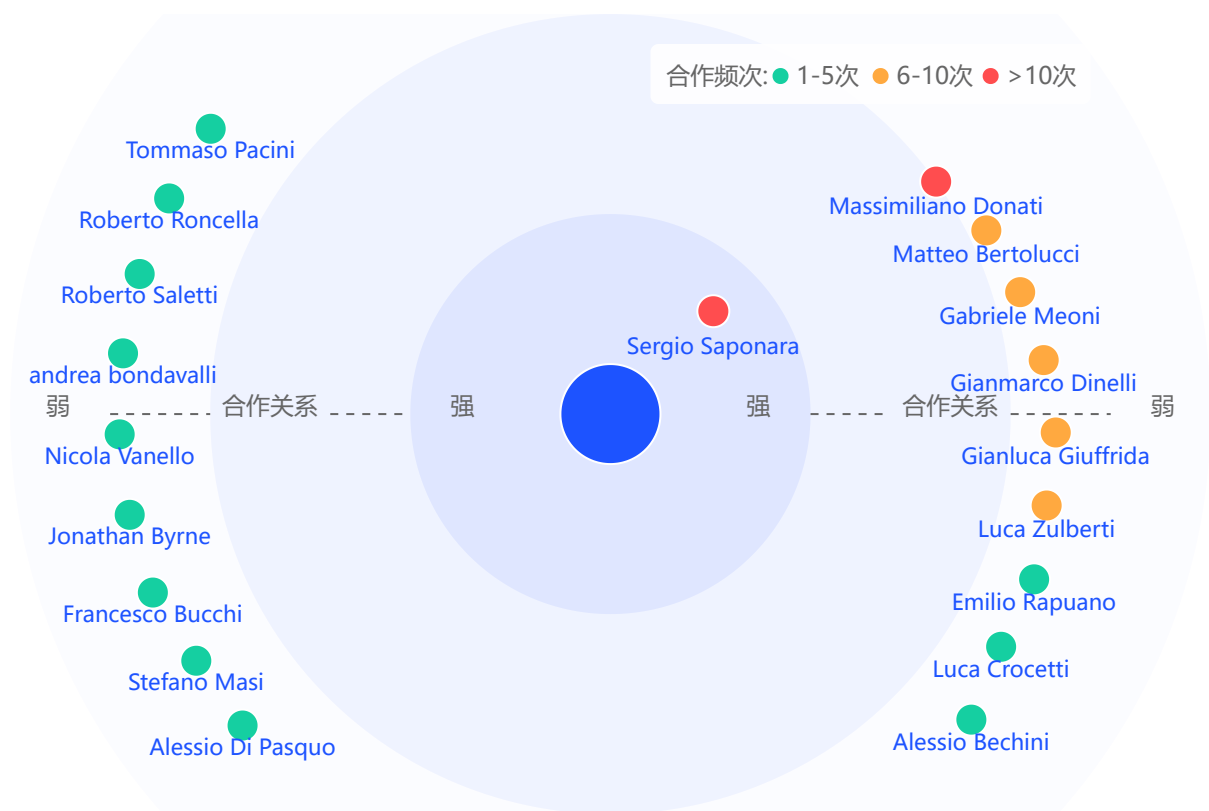
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